

- ### SECTION-D
- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain with neat sketch the working principle and constructional details of radiations pyrometer. Also write its advantages and disadvantages.
- Q.37 Explain with neat diagram feedback or closed control system Also write its applications and advantages & disadvantages.
- Q.38 Explain with block diagram representation the concept and components of automatic control system.

(4) 180562/120562/030544

Q.1 Ultrasonic level measurement works on the principle of generating.

a) Light b) Sound

c) both d) None

Q.2 The range of air purge system is determined by the

a) Tank height b) Length of tube

c) Air pressure d) None of the above

Q.3 Working principle of radiation pyrometer is based on the

a) Wein's Law

b) Stefan Boltzman Law

c) Kirchoff's Law

d) Seebeck effect

Q.4 The smallest change in the value of a measured variable to which an instrument respond is called:

a) Static Error b) Sensitivity

c) Reproducibility d) Dead Zone

(1) 180562/120562/030544

- Q.5 PH meter has _____ Electrodes.
 a) Zero b) One
 c) Two d) Three
- Q.6 On torr is defined as
 a) One mm hg b) One inch Hg
 c) one atmosphere d) One kilopascal
- Q.7 Hydrostatic pressure level indicator is a
 a) Direct method b) Indirect method
 c) Both A & B d) None of the above
- Q.8 Temperature scales has fixed points :
 a) One b) Two
 c) Three d) Four
- Q.9 A black body is a
 a) good absorber b) good emitter
 c) back absorber d) both A & B
- Q.10 Span is
 a) Low & High value
 b) Low Value
 c) difference between high & low Value
 d) High Value

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Define static characteristics.
 Q.12 Define repeatability.

- Q.13 Define PH.
 Q.14 Define dead zone.
 Q.15 Name any one pyrometer.
 Q.16 Convert 300°C into Kelvin (K) & °F.
 Q.17 Define range
 Q.18 Define resolution.
 Q.19 Define controlled variable
 Q.20 Celsius scale has ice point at _____ °C.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain in brief sight glass used for measurement of liquid.
 Q.22 Explain in brief the radiation level detector used for measurement of level.
 Q.23 Explain working of PH meter in brief.
 Q.24 Describe with neat sketch the working of U tube manometer.
 Q.25 Write different sources of errors in filled type thermometers.
 Q.26 Explain in brief the working of bimetallic thermometer.
 Q.27 Define in brief strip chart recorder.
 Q.28 Explain in brief capacitive pressure transducers used for measurement of pressure.
 Q.29 Define oxygen analyzers in brief.